

1. What is a Digital Supply Chain?

A **supply chain** is the complete process of moving a product from the supplier to the final customer.

For example, buying a laptop online:

Supplier → Manufacturer → Warehouse → Logistics Company → Online Store → Customer

A **digital supply chain** uses technologies such as:

- Cloud computing
- APIs
- IoT sensors
- RFID
- Artificial Intelligence
- RPA
- Real-time tracking
- Data analytics
- Automated warehouses

to make this process **faster, more visible and more automated**.

Traditional supply chain

Imagine a warehouse employee manually checks stock every morning.

"How many laptops do we have?"

They count:

1, 2, 3, 4...

Then update Excel.

This creates a delay and introduces the possibility of human error.

Digital supply chain

An RFID/IoT system can automatically detect inventory.

Product enters warehouse

↓

RFID scanner detects product

↓

Inventory system updates automatically

↓

Dashboard shows current stock

↓

System detects low inventory

↓

Purchase order can be triggered

The organisation can therefore see what is happening **without relying entirely on manual checking.**

2. What is RPA?

Robotic Process Automation

RPA is software that performs repetitive, rule-based computer tasks that humans would otherwise perform manually.

Important point for students:

RPA does not necessarily mean a physical robot.

It is often a **software bot** that interacts with applications in a similar way to a human.

For example, an employee might normally:

1. Open an email.
2. Download an invoice.
3. Open Excel.
4. Copy invoice number.
5. Open accounting system.
6. Enter invoice number.
7. Check amount.
8. Save the record.

An RPA bot can perform these repetitive steps automatically.

3. Real-World Example: Invoice Processing

Imagine a company receives **500 supplier invoices every month**.

Manual process

Supplier sends invoice

↓

Employee opens email

↓

Downloads PDF

↓

Reads invoice

↓

Enters information into accounting system

↓

Checks purchase order

↓

Checks amount

↓

Approves invoice

This could take a significant amount of staff time.

RPA process

Supplier sends invoice

↓

RPA bot detects invoice

↓

Extracts information

↓

Checks purchase order

↓

Compares invoice amount

↓

Updates accounting system

↓

Flags exceptions for human review

Now humans focus on **exceptions and decisions**, rather than repetitive data entry.

Important distinction

RPA is particularly useful when:

- rules are clearly defined
 - tasks are repetitive
 - information comes in predictable formats
 - systems do not already have a suitable integration
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4. RPA Example in an E-Commerce Company

Imagine an online store receives:

1,000 orders per day.

Without automation, employees may need to:

- download orders
- update spreadsheets
- check payment
- update inventory
- create delivery records
- send emails

RPA can automate some of these activities.

Example

Customer places order

↓

RPA checks payment status

↓

RPA updates order management system

↓

RPA updates inventory

↓

RPA generates shipping information

↓

RPA sends confirmation email

↓

Customer receives tracking information

This reduces repetitive administrative work.

5. RPA Does NOT Mean Everything Is Automated

This is an important concept for students.

Suppose an order has a suspicious payment.

The RPA bot can identify:

"Payment requires investigation."

It can send the case to a human employee.

So the workflow becomes:

Automation for normal cases

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Human intervention for exceptions

This is often more practical than trying to automate every decision.

6. Real-Time Inventory Systems

The slide mentions:

RFID and IoT sensor integration for precise stock visibility.

Let's simplify this.

What is RFID?

RFID = Radio Frequency Identification.

An RFID tag can be attached to a product, box or pallet.

The tag contains information that can be detected by an RFID reader.

For example:

Product: Wireless Keyboard

SKU: KB1001

Quantity: 1

When the product passes through an RFID reader, the system can record its movement.

7. RFID Real-World Example: Retail Store

Imagine a clothing store with:

10,000 items

Traditional inventory checking might involve employees manually counting products.

With RFID:

New shirt arrives

↓

RFID tag scanned

↓

Inventory increases from:

20 → 21

When sold:

21 → 20

The inventory system can therefore provide much more current information.

Example

A customer searches online:

"Nike running shoes – Size 42"

The system can check inventory at different stores or warehouses.

It may show:

Store A: 3 units

Store B: 0 units

Warehouse: 15 units

This gives the organisation better visibility of stock.

8. IoT in Supply Chains

What is IoT?

IoT = Internet of Things.

It means physical devices or sensors can:

collect data → transmit data → allow systems to analyse/respond to it.

Example: Refrigerated food

Imagine a company transporting:

Fresh seafood

The truck must maintain a specific temperature.

An IoT temperature sensor continuously measures the temperature.

For example:

10:00 AM → 2°C

11:00 AM → 3°C

12:00 PM → 8°C ⚠

The system can automatically alert the logistics manager.

Without IoT, the company might only discover the problem when the shipment arrives.

9. IoT Example: Pharmaceutical Supply Chain

Some medicines require controlled temperatures during transportation.

A digital supply chain could use:

Temperature sensor

↓

Internet connection

↓

Cloud platform

↓

Real-time dashboard

↓

Alert

For example:

⚠ Temperature exceeded acceptable range.

The logistics team can investigate immediately.

This is much more useful than simply discovering the problem after delivery.

10. RPA vs IoT vs RFID

Students often confuse these technologies.

Technology	Main Purpose	Example
RPA	Automates repetitive computer tasks	Automatically process invoices
RFID	Identifies/tracks tagged objects	Track products in warehouse
IoT	Collects and transmits physical-world data	Monitor truck temperature
API	Allows systems to communicate	E-commerce system sends order to courier
AI	Analyses data and supports predictions/decisions	Predict future demand

A simple way to remember:

RPA = Do

RFID = Identify

IoT = Sense

API = Connect

AI = Analyse/Predict

11. Connected Supply Ecosystems

The slide also mentions:

Direct API connectivity between suppliers, logistics providers and end customers.

This means different companies' systems can communicate electronically.

Traditional approach

An online store receives an order.

An employee:

1. Downloads order.
2. Opens courier website.
3. Enters customer address.
4. Enters parcel information.
5. Creates delivery order.
6. Copies tracking number.
7. Updates the online store.

Lots of manual work.

12. API-Based Supply Chain

With APIs:

Online Store

↓

API

↓

Warehouse Management System

↓

API

↓

Courier System

↓

API

↓

Customer Tracking

The systems communicate automatically.

Example

A customer purchases:

Laptop – \$1,500

The e-commerce system automatically sends:

Order ID: ORD10025

Product: Laptop

Quantity: 1

Address: Customer address

Delivery option: Express

to the warehouse.

The warehouse system confirms:

Order packed.

The courier system receives the shipping information.

The customer automatically receives:

Your order has been shipped. Tracking number: XXXXX

No employee needs to manually copy the information between systems.

13. Real-World Example: Amazon-Style Fulfilment

Consider a large e-commerce company.

Step 1 – Customer orders

Customer purchases:

Wireless headphones

Step 2 – Payment

Payment system confirms the transaction.

Step 3 – Inventory

System identifies:

Warehouse Singapore – 25 units

Step 4 – Warehouse

Warehouse system creates a picking task.

Step 5 – Product movement

Barcode/RFID identifies the product.

Step 6 – Packing

The product is packed.

Step 7 – Logistics

Courier receives delivery information through system integration/API.

Step 8 – Tracking

Customer receives real-time updates.

Step 9 – Delivery

Customer receives the package.

Step 10 – Data

The company records:

- delivery time
- customer location
- product demand
- returns
- inventory levels
- customer behaviour

This information can be used to improve future operations.

14. Putting Everything Together

The most important concept in this LO is that these technologies **work together**.

Consider an online grocery company.

Customer

Orders:

5 bottles of milk

Digital Supply Chain

The system receives the order.

↓

API

Order is sent to the warehouse.

↓

RFID

System identifies where the milk is located.

↓

IoT

Refrigerated storage monitors temperature.

↓

RPA

Software automatically creates the picking and delivery records.

↓

Warehouse

Staff/robots pick the products.

↓

Logistics API

Delivery company receives the order.

↓

IoT/GPS

Delivery vehicle location is tracked.

↓

Customer

Receives:

Your order will arrive in approximately 15 minutes.

This is an **end-to-end digital supply chain**.

15. What Happens Without Digitalisation?

Ask students to imagine this scenario:

A company sells 10,000 products online every day but manages its supply chain using Excel, emails and phone calls.

The workflow might be:

Customer order

↓

Email

↓

Staff checks Excel

↓

Phone warehouse

↓

Warehouse checks stock

↓

Calls Sales

↓

Sales emails Finance

↓

Finance checks payment

↓

Emails Warehouse

↓

Warehouse prepares order

↓

Staff calls courier

↓

Courier collects order

↓

Sales emails customer

Problems

Students should identify:

- ✗ Waiting time
 - ✗ Duplicate data entry
 - ✗ Human errors
 - ✗ Poor visibility
 - ✗ Delayed customer updates
 - ✗ Manual work
 - ✗ Difficult inventory management
 - ✗ Poor coordination between departments
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16. Digitalised Version

The same company could transform the process:

Customer Order

↓

E-Commerce Platform

↓

Payment Gateway

↓

Inventory System

↓

RPA

↓

Warehouse Management System

↓

RFID/Barcode

↓

Packing

↓

Courier API

↓

GPS/IoT Tracking

↓

Customer

The information flows automatically.